SOFTWARE ENGINEERING · MINI PROJECT

ReturnLoop

A returns and reverse-logistics centre for an e-commerce business: request, pickup, quality check, grading, refurbish / restock / scrap, refund.



PRESENTED BY 03

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FLASK · SQLITE · THREE.JS

What we'll cover

02

03 Project planning & management

04 Functional requirements

05 Non-functional requirements

06 Use case diagram

07 Class diagram

08 Data flow diagram, level 0

09 Data flow diagram, level 1

10 Activity diagram

11 Sequence diagram

12 Layered architecture

13 Architecture: from UI to data

Vision, scope and stakeholders

03

Product vision

Give an online store one place to run every return, from the customer's doorstep to a graded item and a correct refund, with every step on record.

Objectives

- Enforce return rules automatically
- Grade items A / B / C and route them
- Compute refunds exactly, once
- Live tracking for the customer

In scope

- 6 roles, 13-status workflow
- Pickup booking, 2 attempts
- QC checklist, fraud check
- Refurbish & re-inspection
- Refund + inventory log
- Admin dashboard & flags

Out of scope

- Real payments / payment gateway
- Passwords & real accounts
- Courier API integration

Stakeholders

Customer	returns items, tracks refund
Pickup agent	collects parcels
Warehouse	checks parcels in
QC inspector	grades items
Refurbisher	repairs grade B
Admin	monitors, flags

Team

Chirag Seervi, Suman Singh, Riya Singh

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Incremental delivery in seven increments

Each increment was built, tested with **pytest**, run, and committed to **Git** before the next one started. Definition of done: all tests green and the app runs.

INC A
Core
Models, config, services, unit tests

INC B
Data
Seed data through real services

INC C
Customer
Orders, return, pickup, tracking

INC D
Operations
Agent, warehouse, QC, refurbisher

INC E
Admin
Dashboard, charts, flags

INC F
Experience
Design system, 3D scenes, effects

INC G
Hardening
Edge cases, docs, end-to-end demo

Why incremental?

Business rules first, UI later: the riskiest logic (state machine, refunds) was tested before any screen existed.

Work breakdown

Data model → services → routes → templates → visual design → testing & docs.

Tools

Python, Flask, SQLite, pytest, Git, VS Code, Playwright for screenshot checks.

Risks, and how we handled them

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RISK	IMPACT	MITIGATION IN THE PRODUCT
Invalid status jumps (e.g. refund before QC)	HIGH	Single state machine; every change validated and logged in `status_history`
Refund paid twice	HIGH	Unique `refunds.return_id` + service check + one DB transaction
Double-clicks on submit	MEDIUM	Buttons disable on submit; server rejects the repeat via the state machine
Malicious or huge uploads	MEDIUM	Type, 3 MB size and file-signature checks; random file names
Device without WebGL / slow GPU	LOW	Static fallbacks; particles halve if frame rate < 40 fps
Rules change later (window, fees)	LOW	All rule constants live in `config.py`

QUALITY: 84 AUTOMATED TESTS · STATE MACHINE, REFUNDS, ELIGIBILITY, FLAGS, API, PAGES, FULL FLOW

Functional requirements

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FR01
Role switch
Choose any of 6 roles and a user without a password; each page rejects the wrong role (403).

FR02
View orders
Customer sees delivered items with eligibility, or the reason it can't be returned.

FR03
Eligibility
Return allowed within 7 days of delivery, for returnable products, with no other open return.

FR04
Request return
Reason, comment (≤ 300 chars), optional photo (PNG/JPG/WEBP, ≤ 3 MB).

FR05
Book pickup
Date from tomorrow to +5 days, slot Morning / Afternoon / Evening; reschedule.

FR06
Collect parcel
Agent marks picked up or absent; the 2nd absence cancels the return.

FR07
Receive
Warehouse checks the parcel in; it joins the QC queue.

FR08
Inspect & grade
QC checklist, suggested grade, A/B/C; a mismatched item is rejected as fraud.

FR09
Refurbish
Grade B goes to repair, then re-inspection: pass → restock (70%), fail → scrap.

FR10
Refund
final = max(0, base × grade% − pickup fee), damaged/wrong-item exception, created once.

FR11
Track
Live status (polled every 5 s), history timeline and refund breakdown.

FR12
Dashboard
Totals, reasons and grades charts, frequent-returner flags (> 3 in 30 days).

Non-functional requirements

07

NFR01
Performance
60 fps on a mid laptop, 30+ on phones; pixel ratio capped; 3D pauses off-screen.

NFR02
Reliability
Refund + inventory log + status change in one transaction; no double refunds.

NFR03
Security
Role guard on every blueprint; customers only see their own returns; upload validation.

NFR04
Usability
Works at 360 px wide; tap targets ≥ 44 px; plain-language errors and empty states.

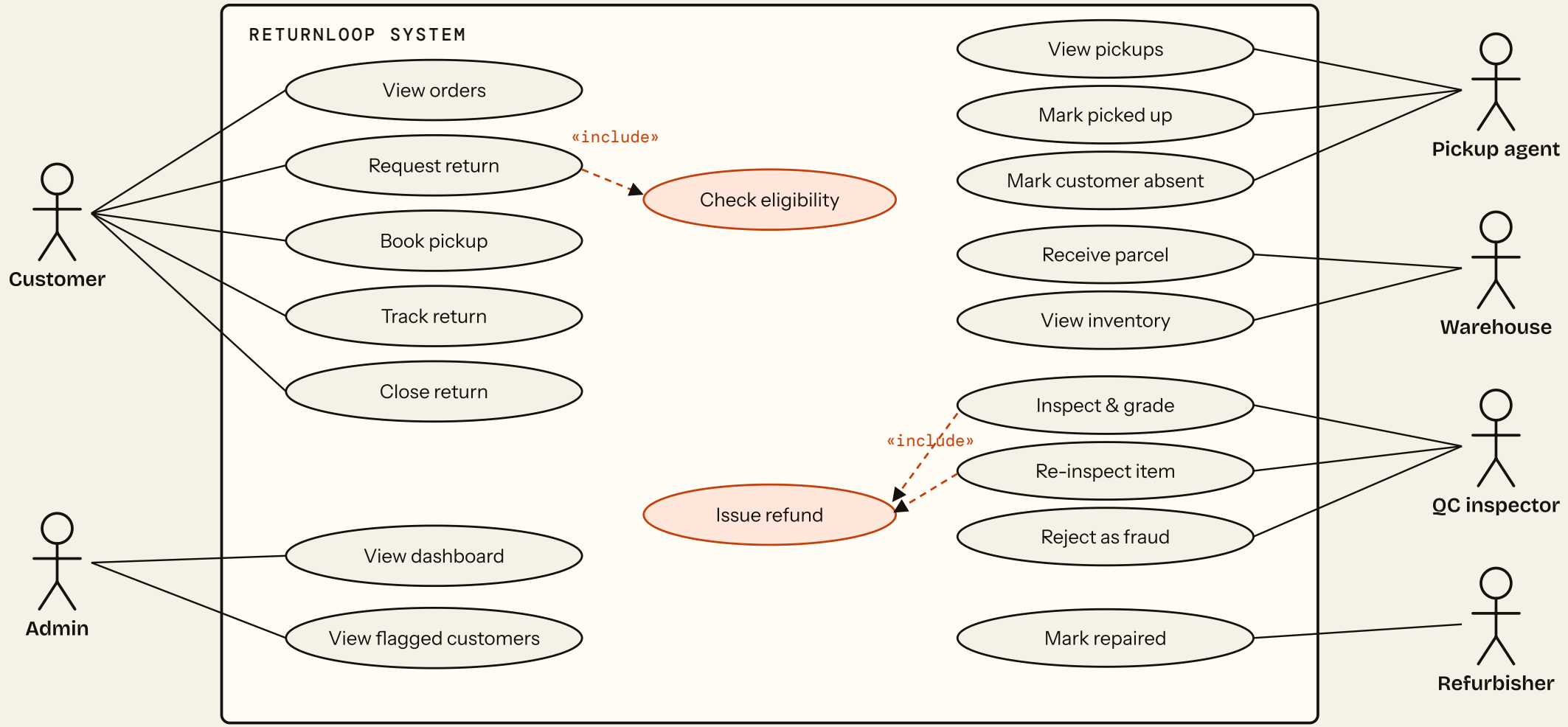
NFR05
Accessibility
WCAG AA contrast, keyboard-only use, visible focus, reduced-motion mode, text on every badge.

NFR06
Maintainability
Logic in services/, rules in config.py, thin routes; 84 pytest tests.

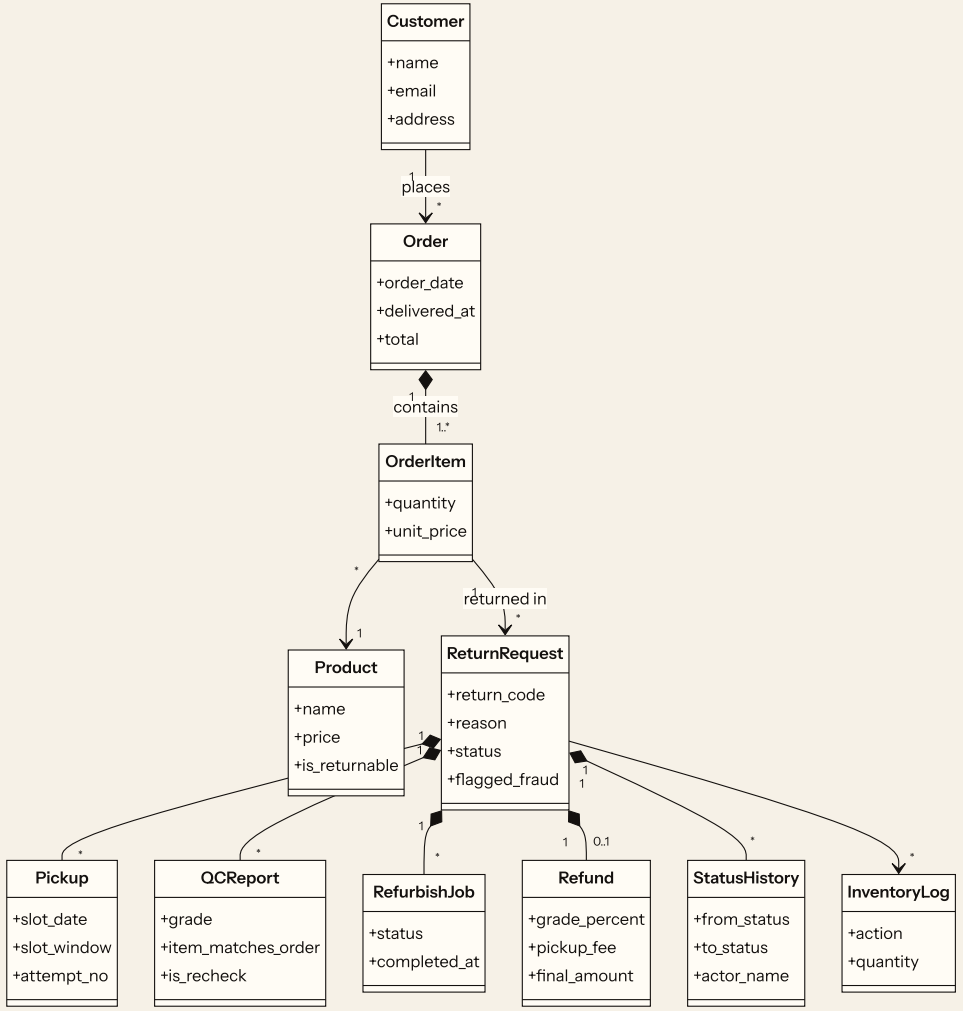
NFR07
Portability
Python 3.10+, SQLite, no build step: install, seed, run in 3 commands.

NFR08
Robustness
Static fallbacks without WebGL or internet; friendly 403 / 404 / 413 pages.

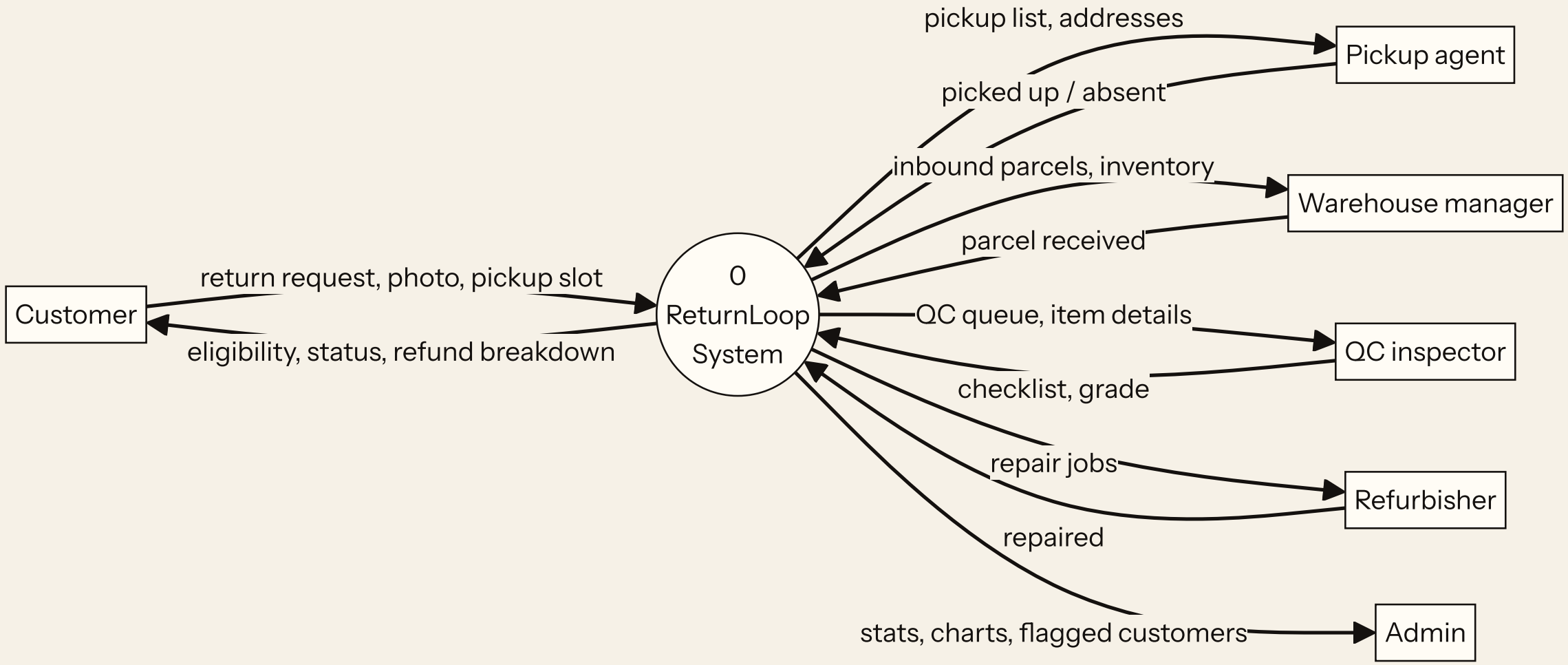
Who does what



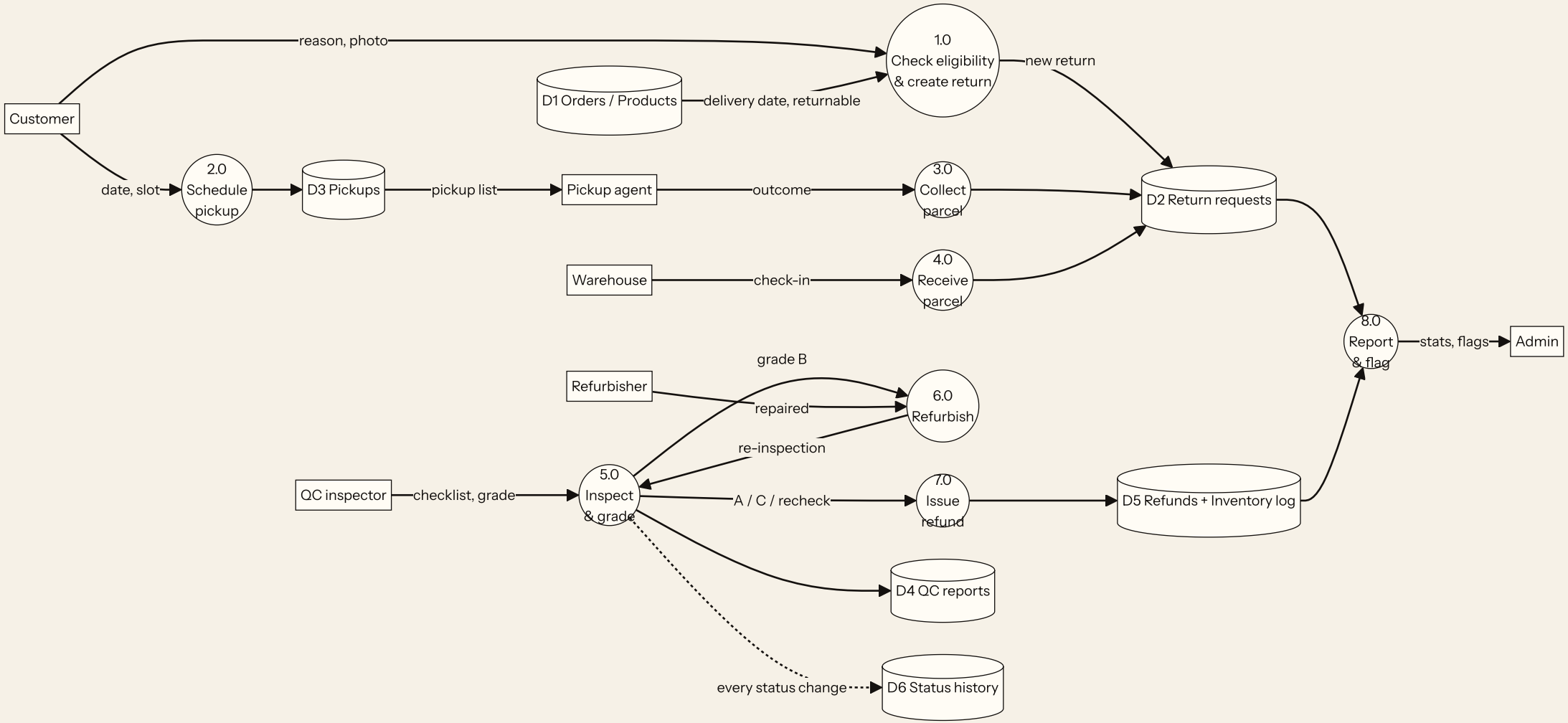
The domain model (models.py)



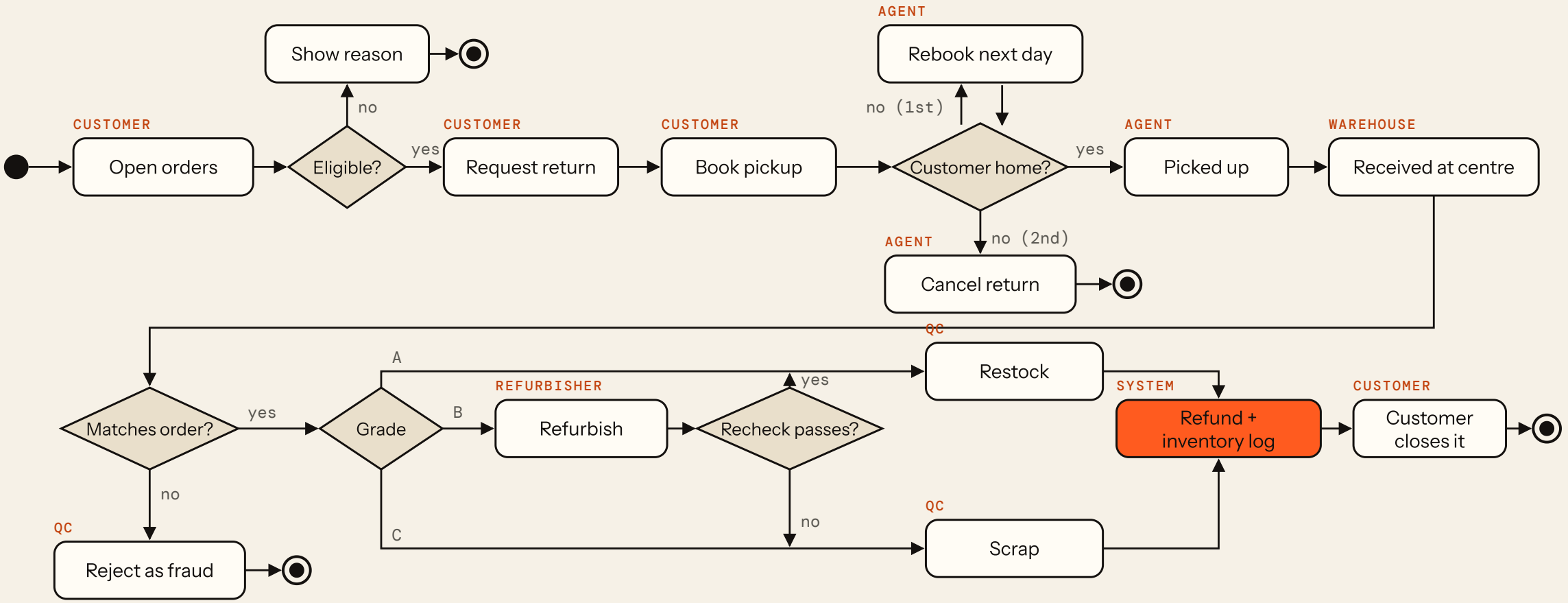
The system and the people around it



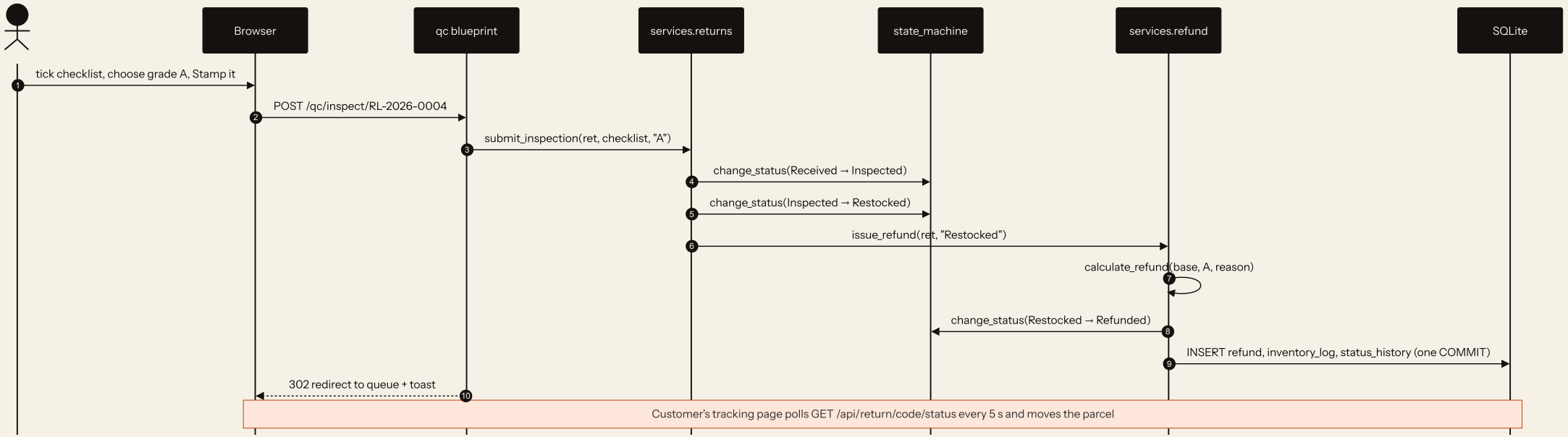
Processes and data stores



One return, start to finish



QC grades a return and the refund is issued



Layered architecture

L1	**Presentation**	Jinja2 templates · main.css · ES modules · Three.js scenes · GSAP · Chart.js	`templates/, static/`
L2	**Controller / routing**	Flask blueprints, one per role + JSON API; role guard on each	`blueprints/`
L3	**Business logic**	Workflow, state machine, eligibility, refund, flags, uploads, stats	`services/`
L4	**Configuration**	Every business rule as a named constant (window, fees, %, limits)	`config.py`
L5	**Data access**	Flask-SQLAlchemy ORM: 11 model classes and their relationships	`models.py`
L6	**Persistence**	SQLite database file, indexes on status, customer, history	`returnloop.db`

RULE: EACH LAYER ONLY TALKS TO THE ONE BELOW · ROUTES NEVER CONTAIN BUSINESS RULES

One click, layer by layer: “Request return”

1 UI
Browser
Form with reason tiles, comment and photo is submitted (button disables itself).

→

2 HTTP
POST /customer/return/new
Multipart request reaches Flask; session holds role + user id.

→

3 Guard
customer blueprint
before_request checks the role is customer, else friendly 403.

4 Validate
services.uploads
Photo type, size ≤ 3 MB and file signature are checked.

→

5 Rules
services.eligibility
7-day window, returnable product, no open return.

→

6 Logic
services.returns
Generates RL-2026-0004, writes the first status_history row.

7 ORM
models.ReturnRequest
SQLAlchemy turns objects into INSERT statements.

→

8 Data
SQLite
Rows committed in one transaction.

→

9 Back to UI
Redirect + Jinja2
Pickup page renders; tracking page later polls the JSON API.

END OF DECK · QUESTIONS?

Thank you.



Chirag Seervi · Suman Singh · Riya Singh

```
pip install -r requirements.txt && python seed.py && python app.py
```